

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	Both reactances cancel or total impedance is R or equivalent (1) $I = \frac{V}{Z} = \frac{V}{R}$ (or equivalent) (1)	1	1		2		
		(iii)	Any correct reactance ($X_C = 10\,105$, $X_L = 10\,556$) (1) Impedance correctly obtained $485\, [\Omega]$ (1) Substitution into $I = \frac{V}{Z}$ ecf on Z (1) Correct answer = $5.15\, \text{mA}$ ecf on Z (1)	1	1 1 1		4	4	
		(iv)	Large decrease in current or $\frac{\Delta I}{\Delta f}$ (1) For small increase in frequency or is large (1)	1	1		2	2	
		(v)	Realising ω also changes OR using / deriving $\frac{1}{R}\sqrt{\frac{L}{C}}$ (1) Correctly linking valid equation to obtain yes OR simply yes e.g.in $\frac{\omega L}{R}$ ω decreases so yes (1)			2	2		
			Question 7 total	6	9	5	20	10	0